SOLUTIONS

DIGITAL SIGNAL PROCESSING

1. a) Compare and contrast FIR and IIR filters. [4]

Solution:
Explanation of structural differences, the role of feedback;
Explanations in terms of poles and zeros;
Stability considerations;
Computational complexity.

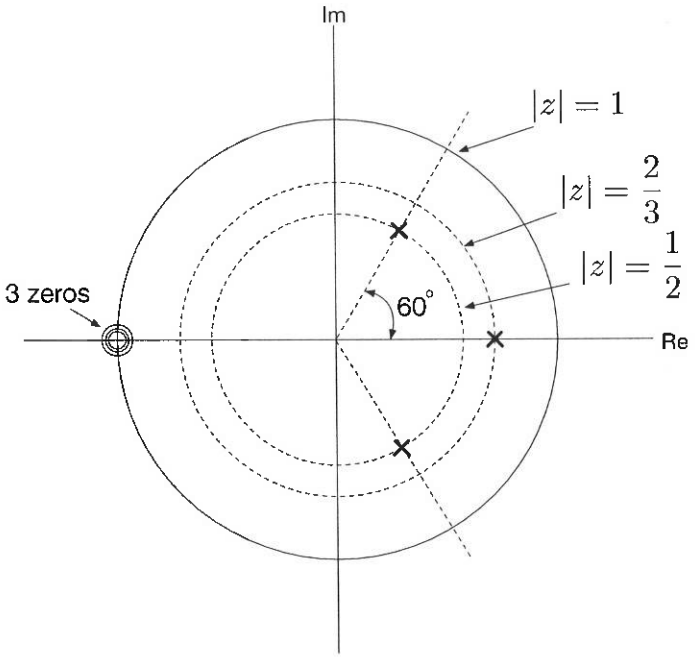


Figure 1.1 Pole-zero diagram

b) The pole-zero plot of a digitial filter is given in Fig. 1.1. The gain of this filter at d.c. is 1.

i) Find the system function of this filter in the form

$$H(z) = A\left[\frac{(1+a_1z^{-1})(1+b_1z^{-1}+b_2z^{-2})}{(1+c_1z^{-1})(1+d_1z^{-1}+d_2z^{-2})}\right]$$

giving the values of the constant coefficients A, a_1, b_1, b_2, c_1, d_1, d_2.

[8]

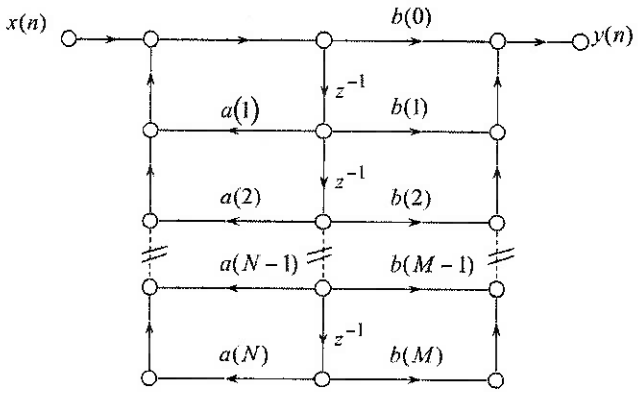
Solution:

$$H(z) = A\left[\frac{(z+1)^3}{(z-\frac{2}{3})(z-(\frac{1}{4}+j\frac{\sqrt{3}}{4})(z-(\frac{1}{4}-j\frac{\sqrt{3}}{4}))}\right]$$
$$= A\left[\frac{(1+z^{-1})(1+2z^{-1}+z^{-2})}{(1-\frac{2}{3}z^{-1})(1-\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2})}\right]$$

Therefore $a_1 = 1$, $b_1 = 2$, $b_2 = 1$, $c_1 = -\frac{2}{3}$, $d_1 = -\frac{1}{2}$, $d_2 = \frac{1}{4}$. For unit gain at d.c. we required $A = 32$.

c) Draw a signal flow graph for the filter in Direct Form II. [4]

Solution:



d) Discuss the impact on the filter's pole-zero plot that would result if the coefficient d_2 was found to be erroneously increased by 10%. [4]

Solution: The impact of an error in the denominator coefficients may be more significant than an error in the numerator coefficients due to error feedback arising in the recursive structure.
The increase from $d_2 = 0.25$ to $d_2' = 0.275$ moves the roots in polynomial d in modulus from 0.5 to 0.52 and in angle from ± 1.05 to ± 1.07.

2. a) Consider the system function given by

$$H(z) = \sum_{k=0}^{\infty} b_k z^{-k}.$$

Describe the characteristics of the filter represented by this system function and state the requirements that must be satisfied for the filter to be causal and stable. [4]

Solution:
This is a nonrecursive (MA) filter. It might be described as 'FIR' but note the semi-infinite support.
In general a system $H(z)$ is stable if its ROC includes the unit circle in the z-plane. The ROC for this filter structure excludes $z = 0$ and therefore includes the unit circle. Accordingly, nonrecursive systems such as this cannot be unstable. Equally $H(z)$ is stable if its inverse z-transform $h(n)$ satisfies

$$\sum_{-\infty}^{\infty} |h(n)| < \infty.$$

A right-sided impulse response indicates causality.

b) Explain the key properties of a linear phase FIR filter. Include discussion in terms of the filter coefficients and in terms of the z-domain. Include relevant sketches to illustrate your explanation. [6]

Solution:
The filter coefficients satisfy $h(n) = h(N-1-n)$. The zeros in the z-domain occur in mirror image pairs such that if a zero exists at z_0 then another zero also exists at $\frac{1}{z_0^*}$. Sketches are required for the filter coefficient symmetry and an example z-plot plot.

c) Show that the frequency response of a linear phase FIR filter with N coefficients, N even, can be written as

$$H(e^{j\omega}) = e^{j\lambda} \sum_{n=0}^{M-1} 2h(n)\cos(\gamma)$$

and give expressions for M, λ and γ. [5]

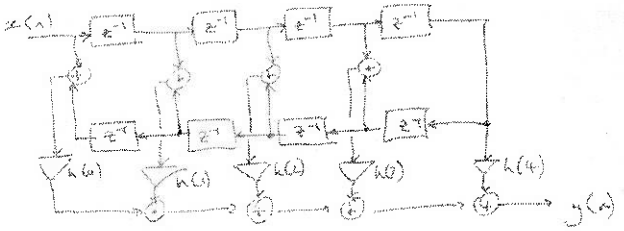
Solution:
This follows from

$$H(z) = \sum_{n=0}^{N/2-1} h(n)\left[z^{-n} + z^{-(N-1-n)}\right]$$

so that $M = N/2$, $\lambda = -\omega[(N-1)/2]$, $\gamma = \omega(n - \frac{N-1}{2})$.

d) Draw and label the signal flow diagram of the efficient implementation of a linear phase FIR filter for $N = 9$. [5]

Solution:



3. Consider an LTI system operating with sampling frequency 10 kHz and system function

$$H(z) = z^{-1}.$$

a) Draw labelled sketch graphs of the magnitude and phase of the frequency response of this system. [2]

Solution: The sketch graphs should clearly show a gain of 1 or 0 dB across frequency with a corresponding phase of $-\omega$, linear with frequency. Axes must be correctly labelled to obtain the marks.

b) Write down the general expression for group delay and give the units of group delay. Determine the group delay of $H(z)$. [3]

Solution:
Group delay $= -\frac{d\phi}{d\omega}$.
Units of time (s). Samples would also be acceptable.
Group delay of $H(z)$ is 0.1 ms.

c) Consider the vector

$$\mathbf{x} = [x(0)\ x(1)\ \cdots\ x(N-1)]^T$$

the elements of which are samples of the input signal and ^{T} indicates the transpose operation. Further consider the vector

$$\mathbf{y} = [y(0)\ y(1)\ \cdots\ y(N-1)]^T$$

the elements of which are samples of the output signal.

i) Given that the processing due to $H(z)$ is formulated as a matrix operation $\mathbf{y} = \mathbf{H}\mathbf{x}$, write out the elements of $\mathbf{H}$. Use the notation $\cdots$ to account for an undefined value for N. [4]

Solution:

$$\mathbf{H} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \ddots & & & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$$

ii) Now write out the elements of $\mathbf{H}_c$ representing the processing due to $H_c(z)$ in matrix form where $H_c(z) = H(z)$ except that any shifts are performed circularly with modulo N. [3]

Solution:

$$\mathbf{H} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & 1 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \ddots & & & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}$$

iii) Given that the DFT is formulated as a matrix operation $\mathbf{y} = \mathbf{F}\mathbf{x}$, write out the elements of $\mathbf{F}$. [3]

Solution:

$$\mathbf{F} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & e^{-j2\pi/N} & e^{-j2.2\pi/N} & \cdots & e^{-j(N-1).2\pi/N} \\ 1 & e^{-j2.2\pi/N} & e^{-j4.2\pi/N} & \cdots & e^{-j2(N-1).2\pi/N} \\ \vdots & \ddots & & & \vdots \\ 1 & e^{-j(N-1).2\pi/N} & e^{-j2((N-1).2\pi/N} & \cdots & e^{-j(N-1)(N-1).2\pi/N} \end{bmatrix}$$

d) Consider writing the processing due to $H_c(z)$ in matrix form so that

$$\mathbf{H}_c = \mathbf{F}^{-1}\mathbf{A}\mathbf{F}.$$

Find the elements of the matrix $\mathbf{A}$. Comment on the significance of this formulation and state any characteristic properties of the rows, columns and diagonals of $\mathbf{F}$ and $\mathbf{A}$ in this context. Discuss briefly the role of normalization in the DFT. [5]

Solution:

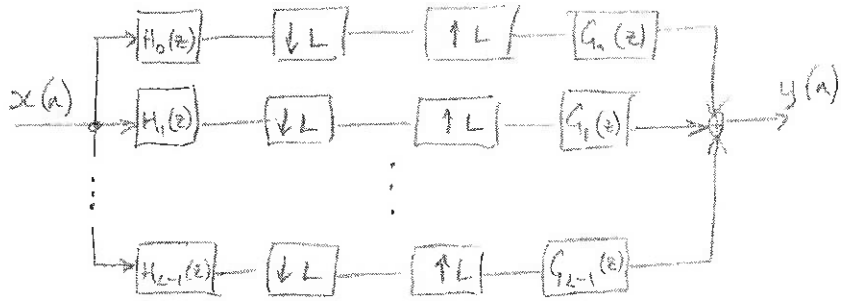
$$\mathbf{A} = N \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & e^{-j2\pi/N} & 0 & \cdots & 0 \\ 0 & 0 & e^{-j2.2\pi/N} & \cdots & 0 \\ \vdots & \ddots & & & \vdots \\ 0 & 0 & 0 & \cdots & e^{-j(N-1).2\pi/N} \end{bmatrix}$$

This formulation shows that the DFT diagonalizes the unit circular delay operator so that the columns of $\mathbf{F}$ are eigenvectors and the diagonal elements of $\mathbf{A}$ are the corresponding eigenvalues. The normalization embeds the term $1/N$ usually in the inverse transform to achieve the unitary property in $\mathbf{F}$.

4. A particular multirate digital signal processing system is formed from an analysis filter bank followed by a synthesis filter bank. The analysis and synthesis filter banks both contain L filters indexed $k = 0, 1, \ldots, L-1$ and are denoted respectively $H_k(z)$ and $G_k(z)$.

a) Draw a block diagram of this multirate system and label the input signal $x(n)$, the output signal $y(n)$ and the filters $H_k(z)$ and $G_k(z)$ for arbitrary L. [4]

Solution:

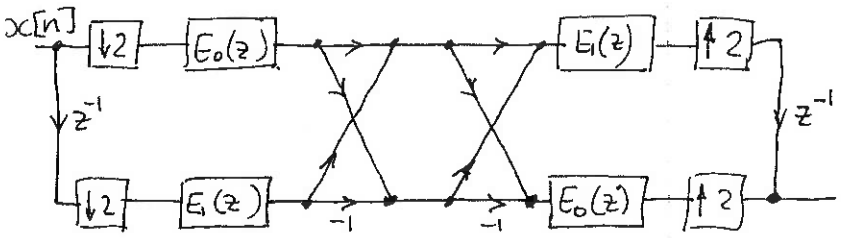


b) Discuss the advantages and disadvantages of polyphase implementations of filter banks. [2]

Solution:
Polyphase filter banks benefit from efficiency in that all filter computations can be performed at the lowest sampling rate in the system.

c) Now draw a fully labelled block diagram of the multirate system of a) with $L = 2$ implemented in Type 1 polyphase form, where the polyphase filters are denoted $E_{k\ell}(z)$ for phase ℓ. [3]

Solution:
Each filter in part a) is replaced with it's polyphase representation. 2 marks if the direct substitution is made. 3 marks if an efficient implementation is provided such as



d) Find the relationship between the coefficients of the filter $E_{k\ell}(z)$ and the coefficients of the filter $H_k(z)$. [3]

Solution:
For the kth filter, the coefficients of $E_{k\ell}(z)$ are a subset of the coefficients of $H(z)$ obtained by subsampling such that

$$e_{k\ell} = h_{kp} \quad p = \ell L$$

e) Show that the expression for $H_k(z)$ can be written in matrix form as

$$\mathbf{h}(z) = \mathbf{E}(z^L)\mathbf{e}(z)$$

and give the definitions of $\mathbf{h}(z)$, $\mathbf{E}(z)$ and $\mathbf{e}(z)$, where $\mathbf{E}(z)$ is called the polyphase component matrix of the analysis filter bank. [4]

Solution:

$$\mathbf{h}(z) = \mathbf{E}(z^L)\mathbf{e}(z)$$

$$\mathbf{h}(z) = [H_0(z)\ H_1(z)\ \ldots\ H_{L-1}(z)]^T$$
$$\mathbf{e}(z) = [1\ z^{-1}\ \ldots\ z^{-(L-1)}]^T$$
$$\mathbf{E}(z) = \begin{bmatrix} E_{00}(z) & E_{01}(z) & \ldots & E_{0,L-1}(z) \\ E_{10}(z) & E_{11}(z) & \ldots & E_{1,L-1}(z) \\ \vdots & \vdots & \ddots & \vdots \\ E_{L-1,0}(z) & E_{L-1,1}(z) & \ldots & E_{L-1,L-1}(z) \end{bmatrix}$$

f) Consider the matrix $\mathbf{R}(z)$ which is the polyphase component matrix of the synthesis filter bank. Define what is meant by *perfect reconstruction* in this context and deduce the condition on $\mathbf{R}(z)$ for perfect reconstruction. [4]

Solution:

Perfect reconstruction:

- Alias cancellation
- No amplitude distortion except for a multiplicative arbitrary gain.
- No phase distortion except for an arbitrary delay.

$$\hat{X}(z) = cz^{-n_0}X(z)$$

for scalar constants c, n_0. In matrix form

$$\mathbf{R}(z)\mathbf{E}(z) = \mathbf{I}.$$